

B.Tech. Degree III Semester Regular/Supplementary Examination January 2023

ME 19-205-0304 FLUID MECHANICS

(2019 Scheme)

Time: 3 Hours

Maximum Marks: 60

Course Outcomes

On successful completion of the course, the students will be able to:

- CO1: Remembering the basic laws such as pascals law, Newton's law of motions, different classification of fluids, Eulerian and Lagrangian approach.
- CO2: Understand the basis of fluid statics, manometry, forces on submerged bodies, stability of bodies in fluid, fluid dynamics, working of flow measuring devices, flow through pipes, stream & Potential functions elementary flows, formation of Boundary layers and Couette flow.
- CO3: Explore the applications of the hydrostatic laws, Continuity and Bernoulli's equations, stream lines, laminar and turbulent flows through pipes and hydraulic coefficients.
- CO4: Apply the knowledge of two-dimensional incompressible flows, stream & potential functions, concept of the formation of superimposed flows, Magnus effect.
- CO5: Apply the concept of boundary layers, Boundary layer equations, skin friction drag in flows.
- Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate, L6 – Create
- PO – Programme Outcome

PART A

(Answer ALL questions)

- | | (8 × 3 = 24) | Marks | BL | CO | PO |
|---|--------------|-------|----|----|-------|
| I. (a) Derive an expression for the force exerted on a submerged vertical plane surface by the static liquid and locate the position of centre of pressure. | 3 | 3 | L2 | 1 | 1 |
| (b) Describe the concept of Eulerian and Lagrangian approach with a simple example. | 3 | 3 | L1 | 1 | 1 |
| (c) Explain with a line sketch the working of an Venturi meter. | 3 | 3 | L2 | 2 | 1 |
| (d) In a possible flow field, the velocity distributions are $u = ?$, $v = ax^3 - by^2 + cz^2$ and $w = bx^3 - cy^2 + az^2$. Determine the missing component. | 3 | 3 | L3 | 3 | 1,2,3 |
| (e) Differentiate stream lines, streak lines and path lines. Write down the equations of each line in an x-y plane. | 3 | 3 | L2 | 2 | 1,2 |
| (f) Define Strouhal number, Reynolds number and Froude Number. | 3 | 3 | L2 | 2 | 1,12 |
| (g) What is Couette flow? Plot the velocity and shear stress distribution within the flow field. | 3 | 3 | L2 | 2 | 1,2 |
| (h) What is meant by hydrodynamic boundary layer? | 3 | 3 | L2 | 2 | 1,12 |

PART B

(4 × 12 = 48)

- | | | | | | |
|--|---|---|----|---|-----------|
| II. (a) Determine the viscosity of a liquid having kinematic viscosity 6.2 stokes and specific gravity 1.87. | 3 | 3 | L3 | 1 | 1,2,3 |
| (b) Obtain the phenomenon of capillarity. Also obtain an expression for capillary rise of a liquid. | 3 | 3 | L2 | 1 | 1,2 |
| (c) A block of wood of specific gravity 0.72 floats in water. Determine the meta-centric height of the block if its size is 2 m × 1 m × 0.8 m. | 6 | 6 | L3 | 3 | 1,2,3, 12 |

OR

(P.T.O.)

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		Marks	BL	CO	PO
III.	(a) Determine the equation of stream line for two -dimensional flow described by $u = xe^{-kt}$ and $v = y$.	3	L3	2	1,2
	(b) If the velocity components for a two- dimensional flow are given by $V = \left(10 + \frac{20x}{x^2 + y^2}\right)i + \left(\frac{20y}{x^2 + y^2}\right)j$. Find the accelerations in x, y, z directions. Also, the total acceleration.	5	L3	2	1,2,3
	(c) Determine the circulation of $V = 4x^2i + 2yzj - (4y^2 + z^2)k$, about a closed triangle which has its vertices (0,0,0), (0,4,0) and (4,0,0).	4	L3	2	1,2,3
IV.	(a) Derive an expression for the time taken to empty a cylindrical tank by a small orifice at the bottom.	3	L2	3	1,2
	(b) With a schematic sketch explain hydraulic coefficients.	3	L2	2	1
	(c) Water discharge at the rate of 100 litres/s through a 120 mm diameter vertical orifice placed under a constant head of 10 meters. A point, on the jet, measured from the vena-contracta of the jet has coordinates 4.6 meters horizontal and 0.55 meters vertical. Find the coefficient C_v, C_c and C_d of the orifice.	6	L3	3	1,2,12
OR					
V.	(a) What is meant by local losses in pipes? Derive an expression for the major losses in a horizontal piped of diameter 'D' as function of friction coefficient 'f'. Indicate the assumptions used.	6	L2	2	1,2,12
	(b) A liquid with a viscosity $\mu = 0.40 \text{ N.s/m}^2$ and density $\rho = 900 \text{ kg/m}^3$ flows in a pipe of diameter $D = 0.020 \text{ m}$.	6	L4	3	1,2,12
	(i) What pressure drop is needed to produce a flow rate of 0.02 liters/sec if the pipe is horizontal with $x_1 = 0$ and $x_2 = 10 \text{ m}$?				
	(ii) How steep a hill, θ , must the pipe be on if the oil is to flow through the pipe at the same rate as in part (i), but with $P_1 = P_2$?				
	(iii) If $P_1 = 200 \text{ kPa}$, what is the pressure at section $x_1 = 5 \text{ m}$, where x is measured along the pipe?				
VI.	(a) State the definitions of potential and stream functions for two - dimensional flow. What conditions have to be satisfied so that they can exist?	5	L2	2	1,2,3
	(b) The velocity field is given by $u = x^2 - y^2 + x$ and $v = -(2xy + y)$	7	L3	4	1,2,3,12
	(i) Check the possibility of the flow.				
	(ii) Show that these functions represent a possible case of irrotational flows.				
	(iii) Obtain an expression for stream function and potential function.				

OR

(Continued)

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		Marks	BL	CO	PO
VII.	(a) What is a dipole? Draw stream lines and potential lines in a dipole field.	3	L3	4	1
	(b) Explain Magnus effect. Draw the flow pattern and wake in the case of a table tennis ball spinning in the forward direction when moving forward.	4	L3	4	1,2,12
	(c) Given the stream function $\varphi = U_{\infty}y \left(1 - \frac{R^2}{x^2 + y^2} \right)$.	5	L4	4	1,2,12
	(i) Sketch the stream lines for $x^2 + y^2 \geq R^2$.				
	(ii) Determine the pressure distribution on the contour $\varphi = 0$.				
	(iii) Determine the time for a particle to move from the point $x = -3R, y = 0$ to the point $x = 2R, y = 0$.				
VIII.	Derive Navier – Stokes equations. State the assumptions used.	12	L2	5	1,3,12
	OR				
IX.	(a) Derive Prandtl boundary layer equations.	6	L2	5	1,3
	(b) What do you mean by separation of boundary layers? What is the effect of pressure gradient on boundary layer separation? What are the different methods of preventing the separation of boundary layers?	6	L2	5	1,3

Bloom's Taxonomy Levels

L1= 2.5%, L2 = 51.6%, L3 = 36.7%, L4 = 9.2%.
